DA01 Open a notebook and read your first table

Week 1, Meet the data | Session 1 of 24 | Python for Data Analysis | New library: pandas

Today's question. What does one week of sales in a small shop look like?

Objective. By the end: you can open, run, save and reopen a notebook, and read a table of shop sales into it.

Before this session: START_HERE.pdf. No subject knowledge is assumed. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
notebook	A page where you write instructions for the computer and see the answers under them.	An exercise book that answers back.
cell	One box in a notebook. A text cell explains. A code cell runs.	One paragraph on a page of the exercise book.
run	Tell the computer to carry out the code in a cell. Click the cell and press Shift and Enter.	Handing in a question and waiting for the answer.
output	What a cell shows after it runs.	The answer written under the question.
file	A saved piece of work with a name, such as camp_shop.csv.	A page stored in a folder.
table	Information arranged in rows and columns.	The sales page of a shop register.

Block A. Run your first cell

Read AI_USE_POLICY.pdf with your instructor now. It explains when AI tools are used in this camp and when they are not. Then click the code cell below and press Shift and Enter. Python is a language for giving a computer instructions. The word print tells it to show something. The brackets hold what to show.

Predict. Predict the two lines that will appear under the cell. Write your guess on paper first.

Your guess, written before you run it

```
print("Hello, data")
print(3 + 2)
```

Run it, then compare. The output is Hello, data on the first line and 5 on the second. Text sits between matching quotation marks. A number such as 3 is an amount.

Investigate.

1. Why does the first line keep its quotation marks in the code and lose them in the output?

2. What would print(3 + 2) show if the 3 were written as "3"?

Modify. Change Hello, data to your nickname and run the cell again.

Block B. Read the shop table

The file camp_shop.csv holds 60 sales from a practice shop. It has to sit in the same folder as this notebook. The three lines below are ready-made. You will learn how each part works in session 7. For now, read what comes out. The word head means the first rows.

Predict. Predict how many rows the table shows and which columns it has. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
shop = pd.read_csv("camp_shop.csv")
print(shop.head())
```

Run it, then compare. The output shows the first five sales as a table. The columns are sale_id, date, item, units and unit_price_ugx. Sale 1 is one Book at 3000 shillings.

Investigate.

1. How many columns does the table have, and what does each one record?
2. Why does the first row show a sale of only 1 unit while the fourth shows 7?

Modify. Change head() to head(10) and run it again.

Make

In a new code cell show your nickname and the result of 7 + 4. Then show the first three rows of the shop table with head(3). Save the notebook as DA01.ipynb, close it and open it again.

Save your work as DA01. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
SyntaxError: invalid character	A curly quotation mark was pasted in. Python only accepts straight quotation marks.	Delete the quotation marks and type them again from the keyboard.
FileNotFoundError: camp_shop.csv	The notebook cannot find the file because it sits in a different folder.	Move the notebook into the folder that holds camp_shop.csv, or ask the instructor to help.

Exit ticket

I can run a cell, read its output, and open a table of sales in a notebook.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or assessments. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been submitted until you use the route in START_HERE.pdf.